

**BRENNER, Austria**

WMO: 111280

Lat: 47.0072N

Lon: 11.5108E

Elev: 4747

StdP: 12.34

Time Zone: 1.00 (EUC)

Period: 94-08

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |            |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%      | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>7.1 | (c)<br>10.2 | (d)<br>1.4                    | (e)<br>7.0 | (f)<br>10.5 | (g)<br>4.1 | (h)<br>8.1 | (i)<br>13.4 | (j)<br>19.2           | (k)<br>26.3 | (l)<br>15.3 | (m)<br>26.7 | (n)<br>2.8            | (o)<br>90 | (p)<br>0.623 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>17.4            | (c)<br>75.4     | (d)<br>59.3 | (e)<br>72.3 | (f)<br>57.9 | (g)<br>69.3 | (h)<br>56.5 | (i)<br>60.8         | (j)<br>71.9 | (k)<br>59.3 | (l)<br>68.9 | (m)<br>57.9 | (n)<br>66.5 | (o)<br>5.9           | (p)<br>230 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 57.1                            | 83.0 | 62.8 | 55.7 | 78.9 | 61.6 | 54.4 | 75.3 | 60.4 | 29.3          | 72.1 | 28.1 | 69.2 | 27.1 | 67.0 | 66.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 16.0              | 11.6 | 9.6 | 2.1                        | 80.1 | 5.7                | 2.4 | -2.1                                               | 81.8 | -5.4       | 83.2 | -8.6       | 84.5 | -12.8      | 86.3 |
|                   |      |     | 1.7                        | 64.1 | 5.5                | 1.9 | -2.2                                               | 65.5 | -5.4       | 66.6 | -8.5       | 67.7 | -12.5      | 69.1 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 40.9   | 24.7  | 26.8  | 32.4  | 39.1  | 48.4  | 53.9  | 56.9  | 55.9  | 49.6  | 43.2  | 32.1  | 26.7 |  |  |
|                                                                        | DBStd       | 13.48  | 7.06  | 7.31  | 8.00  | 7.41  | 6.48  | 7.39  | 5.95  | 6.03  | 6.42  | 6.68  | 7.50  | 6.82 |  |  |
|                                                                        | HDD50       | 4063   | 785   | 648   | 547   | 333   | 109   | 47    | 14    | 17    | 83    | 221   | 536   | 723  |  |  |
|                                                                        | HDD65       | 8809   | 1250  | 1068  | 1012  | 778   | 515   | 334   | 256   | 287   | 461   | 674   | 986   | 1188 |  |  |
|                                                                        | CDD50       | 738    | 0     | 0     | 0     | 5     | 59    | 165   | 227   | 199   | 72    | 11    | 0     | 0    |  |  |
|                                                                        | CDD65       | 10     | 0     | 0     | 0     | 0     | 0     | 2     | 4     | 4     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH74       | 123    | 0     | 0     | 0     | 0     | 2     | 32    | 48    | 40    | 1     | 0     | 0     | 0    |  |  |
|                                                                        | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | WSAvg       | 3.5    | 3.2   | 3.4   | 3.8   | 3.9   | 4.1   | 3.9   | 3.9   | 3.2   | 3.1   | 3.1   | 3.2   | 2.9  |  |  |
|                                                                        | PrecAvg     | 42.9   | 2.0   | 1.7   | 2.2   | 2.8   | 4.2   | 5.4   | 5.9   | 6.1   | 4.0   | 3.5   | 3.0   | 1.9  |  |  |
| (15) Precipitation                                                     | PrecMax     | 53.8   | 4.1   | 4.6   | 6.3   | 5.3   | 6.8   | 7.6   | 9.6   | 9.6   | 6.1   | 6.9   | 7.6   | 4.1  |  |  |
|                                                                        | PrecMin     | 34.7   | 0.4   | 0.4   | 0.7   | 0.3   | 1.7   | 2.9   | 2.8   | 3.1   | 1.2   | 1.1   | 0.0   | 0.0  |  |  |
|                                                                        | PrecStd     | 5.0    | 0.9   | 1.0   | 1.2   | 1.4   | 1.5   | 1.4   | 1.6   | 1.7   | 1.3   | 1.5   | 1.9   | 1.0  |  |  |
|                                                                        | PrecStd     | 5.0    | 0.9   | 1.0   | 1.2   | 1.4   | 1.5   | 1.4   | 1.6   | 1.7   | 1.3   | 1.5   | 1.9   | 1.0  |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 42.9   | 45.8  | 54.7  | 63.8  | 72.8  | 77.9  | 78.7  | 79.0  | 71.7  | 61.1  | 50.7  | 41.9  |      |  |  |
|                                                                        | 0.4% MCWB   | 36.1   | 37.4  | 42.9  | 49.8  | 56.5  | 60.4  | 61.2  | 61.3  | 57.0  | 51.8  | 46.0  | 36.9  |      |  |  |
|                                                                        | 2% DB       | 37.5   | 41.7  | 50.8  | 58.4  | 68.0  | 74.4  | 75.0  | 74.3  | 66.5  | 57.8  | 46.4  | 38.4  |      |  |  |
|                                                                        | 2% MCWB     | 32.8   | 34.7  | 40.7  | 46.7  | 54.0  | 58.4  | 59.8  | 58.8  | 54.3  | 50.4  | 42.5  | 35.3  |      |  |  |
|                                                                        | 5% DB       | 35.2   | 38.3  | 46.8  | 54.2  | 63.8  | 70.7  | 71.4  | 70.7  | 63.0  | 54.7  | 43.6  | 36.5  |      |  |  |
|                                                                        | 5% MCWB     | 31.9   | 32.7  | 38.4  | 44.1  | 51.5  | 56.9  | 58.2  | 57.1  | 52.6  | 48.4  | 41.1  | 33.5  |      |  |  |
|                                                                        | 10% DB      | 33.2   | 35.9  | 43.2  | 50.5  | 59.8  | 67.1  | 68.1  | 67.0  | 59.7  | 52.0  | 41.6  | 34.7  |      |  |  |
|                                                                        | 10% MCWB    | 31.0   | 31.6  | 36.8  | 42.1  | 49.6  | 55.5  | 56.5  | 56.4  | 51.8  | 47.2  | 39.5  | 32.5  |      |  |  |
|                                                                        | 0.4% WB     | 36.7   | 38.5  | 43.9  | 50.7  | 57.4  | 62.3  | 63.0  | 62.7  | 59.0  | 53.6  | 47.8  | 39.1  |      |  |  |
|                                                                        | 0.4% MCDB   | 40.0   | 44.1  | 52.9  | 63.0  | 71.0  | 74.0  | 74.7  | 75.9  | 67.2  | 58.2  | 49.9  | 40.0  |      |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 34.4   | 35.7  | 41.3  | 47.5  | 54.7  | 59.9  | 60.7  | 60.4  | 56.4  | 51.7  | 43.5  | 36.1  |      |  |  |
|                                                                        | 2% MCDB     | 36.1   | 39.8  | 48.8  | 56.5  | 65.5  | 70.9  | 71.8  | 70.8  | 64.1  | 55.8  | 45.4  | 37.4  |      |  |  |
|                                                                        | 5% WB       | 32.9   | 33.8  | 39.4  | 45.2  | 52.6  | 58.2  | 59.3  | 58.5  | 54.4  | 49.9  | 41.4  | 34.5  |      |  |  |
|                                                                        | 5% MCDB     | 34.4   | 36.9  | 45.2  | 52.2  | 61.8  | 67.9  | 68.7  | 67.1  | 60.2  | 53.5  | 43.3  | 35.8  |      |  |  |
|                                                                        | 10% WB      | 31.3   | 32.2  | 37.5  | 43.0  | 50.6  | 56.5  | 57.9  | 57.1  | 52.7  | 47.8  | 39.7  | 33.0  |      |  |  |
|                                                                        | 10% MCDB    | 32.9   | 35.1  | 41.9  | 48.7  | 57.9  | 65.1  | 65.9  | 64.5  | 58.0  | 51.4  | 41.4  | 34.2  |      |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 9.0    | 11.6  | 14.0  | 14.3  | 16.3  | 17.0  | 17.4  | 15.8  | 15.4  | 12.2  | 8.4   | 8.1   |      |  |  |
|                                                                        | 5% DB       | 9.9    | 14.7  | 19.0  | 21.3  | 23.3  | 23.8  | 22.7  | 22.4  | 20.9  | 15.8  | 10.3  | 8.9   |      |  |  |
|                                                                        | MCWBR       | 7.4    | 9.8   | 10.7  | 10.4  | 10.1  | 9.7   | 9.2   | 9.4   | 9.9   | 9.3   | 7.1   | 6.7   |      |  |  |
|                                                                        | 5% WB       | 8.7    | 12.8  | 17.2  | 19.4  | 20.8  | 21.6  | 20.7  | 19.9  | 18.7  | 13.7  | 8.8   | 7.4   |      |  |  |
|                                                                        | MCWBR       | 7.3    | 9.3   | 10.2  | 9.6   | 9.6   | 9.6   | 9.5   | 9.5   | 9.7   | 8.4   | 6.6   | 6.0   |      |  |  |
|                                                                        | MCWBR       | 7.3    | 9.3   | 10.2  | 9.6   | 9.6   | 9.6   | 9.5   | 9.5   | 9.7   | 8.4   | 6.6   | 6.0   |      |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.240  | 0.264 | 0.303 | 0.346 | 0.360 | 0.368 | 0.365 | 0.359 | 0.329 | 0.305 | 0.273 | 0.241 |      |  |  |
|                                                                        | taud        | 2.371  | 2.306 | 2.277 | 2.305 | 2.339 | 2.357 | 2.361 | 2.397 | 2.466 | 2.507 | 2.513 | 2.437 |      |  |  |
|                                                                        | Ebn at Noon | 277    | 290   | 292   | 288   | 287   | 284   | 283   | 280   | 281   | 272   | 263   | 265   |      |  |  |
|                                                                        | Edh at Noon | 22     | 29    | 35    | 38    | 38    | 38    | 37    | 35    | 29    | 24    | 19    | 18    |      |  |  |
| (44) All-Sky Solar Radiation                                           |             | RadAvg | 469   | 768   | 1132  | 1394  | 1538  | 1741  | 1745  | 1534  | 1165  | 792   | 471   | 388  |  |  |
| (45) All-Sky Solar Radiation                                           |             | RadStd | 53    | 91    | 132   | 188   | 171   | 155   | 148   | 153   | 128   | 96    | 74    | 51   |  |  |

**Historical Trends**

|                              |  | Heating |        |       | Cooling |       |       | Degree-Days |       |       |     |
|------------------------------|--|---------|--------|-------|---------|-------|-------|-------------|-------|-------|-----|
|                              |  | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP | HDD50 | HDD65       | CDD50 | CDD65 |     |
| (46) Station Only            |  | N/A     | N/A    | N/A   | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   |     |
| (47) Regional (91 neighbors) |  | +0.67   | N/A    | N/A   | +1.15   | +0.72 | +0.68 | N/A         | N/A   | +112  | +25 |

Nomenclature: See separate page